

Data splitting, not model capacity, explains reported accuracy in a small image-based ECG arrhythmia classifier: a replication

Oshini Wathsala Muthukumarana Palavinnage¹ · Pramudya Thilakaratne²

¹La Trobe University, Melbourne, Australia · ²NSBM Green University, Homagama, Sri Lanka
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Abstract. Automated arrhythmia classification on the MIT-BIH Arrhythmia Database is frequently evaluated with beats from the same patient present in both training and test sets. This practice is known to inflate reported performance, yet it remains common, and results are often summarised by overall accuracy on a heavily imbalanced label distribution. We replicate this effect on an image-based pipeline. A small convolutional network (approximately 18,000 parameters), taken unchanged from an undergraduate project that originally reported over 80 per cent accuracy, is trained twice on identical beat images; the data-splitting protocol is the only variable. Across five random seeds, a beat-level random split gives 97.2 ± 0.3 per cent accuracy and 0.806 ± 0.029 macro-F1 across the N, S, V and F classes. Under the inter-patient split of de Chazal et al., accuracy falls only to 91.3 ± 0.5 per cent while macro-F1 falls to 0.422 ± 0.008 , a relative reduction of 48 per cent: supraventricular sensitivity drops from 0.563 to 0.011 and fusion sensitivity from 0.483 to 0.003. A classifier that always predicted the normal class would score 89.0 per cent on the same test set. We report per-class metrics and release the full pipeline. This is a replication of an established finding, offered as a controlled demonstration on an image-based pipeline rather than as a new result.

1. Introduction

The MIT-BIH Arrhythmia Database [1,2] is the most widely used benchmark for automated heartbeat classification. Two evaluation paradigms are in use. In the intra-patient (or beat-level) paradigm, beats are pooled across patients and divided at random, so beats recorded from a given patient may appear in both the training and test sets. In the inter-patient paradigm, the division is made by patient, so no patient contributes to both. De Chazal et al. [3] introduced the inter-patient division of the database into the record sets DS1 and DS2 and showed that pooling beats allows a model to memorise patient-specific morphology, producing test performance that does not reflect behaviour on unseen individuals.

This distinction is well documented. A recent systematic review of the field reports that only 7.56 per cent of studies addressed the inter-patient paradigm exclusively, while 89.4 per cent considered only intra-patient schemes, and that across eleven studies using both, average accuracy was 98.39 per cent intra-patient against 90.15 per cent inter-patient [4,5]. Earlier work re-implemented published deep learning models and retested them under the AAMI inter-patient split, reaching the same conclusion [6]. Poor minority-class sensitivity, particularly for the supraventricular (S) and fusion (F) classes, is likewise a recognised limitation [7].

We add no new claim to this literature. Our contribution is a controlled replication on an image-based pipeline, in which beats are rendered as images and classified by a convolutional network. The model is small and was not designed for this study: it is taken unchanged from an undergraduate honours project that reported over 80 per cent accuracy under beat-level splitting. Holding everything constant except the split isolates the protocol's contribution to the reported number, and the per-class breakdown shows where the apparent performance actually comes from.

2. Methods

2.1 Data

We use the MIT-BIH Arrhythmia Database, sampled at 360 Hz. Following AAMI practice we exclude the four records containing paced beats (102, 104, 107, 217), leaving 44 records. Lead MLII is used throughout (records in which the leads are ordered differently are handled by name rather than by index). Beat annotations are mapped to the five AAMI classes: N (normal, left and right bundle branch block, atrial and nodal escape), S (atrial premature, aberrated atrial premature, nodal premature, supraventricular premature), V (premature ventricular contraction, ventricular escape), F (fusion of ventricular and normal) and Q (paced, fusion of paced and normal, unclassifiable).

Each annotated beat is extracted as a window of 90 samples before and 162 samples after the R peak (0.25 s and 0.45 s respectively) and rendered as a 120×120 grayscale image: a single-pixel black trace on a white background, amplitude scaled to the window's own range. This mirrors the image representation used in the original project. The record identifier is retained for every beat, which the original pipeline did not do and which is what makes the inter-patient split possible. The procedure yields 100,694 beats: 90,088 N, 2,781 S, 7,008 V, 802 F and 15 Q.

2.2 Model and training

The network is unchanged from the original project: a 3×3 convolution from 1 to 4 channels, ReLU, 2×2 max pooling, a second 3×3 convolution from 4 to 4 channels, ReLU, 2×2 max pooling, dropout at 0.5, and a single linear layer to five outputs; approximately 18,000 parameters in total. Inputs are scaled to [0,1] and normalised to mean 0.5 and standard deviation 0.5. Training uses stochastic gradient descent with learning rate 0.001, momentum 0.9, batch size 32, unweighted cross-entropy, and five epochs. No tuning, class weighting or resampling is applied, since the aim is to reproduce the original configuration rather than to improve it. Each protocol is run with five random seeds (0–4); results are reported as mean $\pm$ standard deviation.

2.3 Protocols

Protocol A (beat-level). All 100,694 beats are pooled and divided at random into 80 per cent training and 20 per cent test, as in the original project.

Protocol B (inter-patient). The DS1/DS2 division of de Chazal et al. [3] is used: 22 records for training (51,002 beats) and 22 disjoint records for testing (49,692 beats).

2.4 Metrics

We report overall accuracy, and per class the sensitivity (Se), positive predictive value (PPV) and F1. The headline summary is macro-F1 over N, S, V and F. Class Q is reported but excluded from macro-F1, as it contributes 15 beats across the whole dataset. We also state the majority-class baseline, the accuracy obtained by always predicting N, because accuracy on this label distribution is otherwise difficult to interpret.

3. Results

Metric	Protocol A beat-level	Protocol B inter-patient
Overall accuracy	97.18 $\pm$ 0.28 %	91.26 $\pm$ 0.54 %
Majority-class baseline	89.8 %	89.0 %
Macro-F1 (N, S, V, F)	0.806 $\pm$ 0.029	0.422 $\pm$ 0.008
N sensitivity	0.996 $\pm$ 0.002	0.971 $\pm$ 0.008
S sensitivity	0.563 $\pm$ 0.040	0.011 $\pm$ 0.006
V sensitivity	0.878 $\pm$ 0.017	0.734 $\pm$ 0.023
F sensitivity	0.483 $\pm$ 0.099	0.003 $\pm$ 0.000

Table 1. Identical model, hyperparameters and images under both protocols; only the split differs. Mean $\pm$ standard deviation over five seeds.

Class	n	Se	PPV	F1	n	Se	PPV	F1
N	18,092	0.996	0.975	0.985	44,240	0.971	0.938	0.954
S	511	0.563	0.907	0.693	1,837	0.011	0.104	0.020
V	1,371	0.878	0.950	0.912	3,220	0.734	0.696	0.712
F	162	0.483	0.942	0.634	388	0.003	0.004	0.003
Q	3	0.000	0.000	0.000	7	0.000	0.000	0.000

Table 2. Per-class results, means over five seeds. Columns 2–5: Protocol A. Columns 6–9: Protocol B.

Overall accuracy falls by about six percentage points between the two protocols, which on its own reads as a modest degradation. The per-class figures show something different. Supraventricular sensitivity collapses from 0.563 to 0.011, roughly 20 correctly identified beats out of 1,837, and fusion from 0.483 to 0.003, around one beat out of 388. Ventricular sensitivity falls from 0.878 to 0.734, with positive predictive value falling from 0.950 to 0.696. Macro-F1 falls by 48 per cent in relative terms, from 0.806 to 0.422, and the variation between seeds (standard deviation 0.008 under Protocol B) is far smaller than the gap between protocols. Under Protocol B the model exceeds the majority-class baseline by 2.3 percentage points, while detecting almost no abnormal beats outside the ventricular class.

4. Discussion

The two numbers that would ordinarily be reported, 97.2 and 91.3 per cent accuracy, differ by far less than the per-class results do. Both exceed the majority-class baseline, and neither conveys that under Protocol B the classifier identifies about one per cent of supraventricular beats. Overall accuracy is close to uninformative on this label distribution, and its continued use as a headline figure is the mechanism by which protocol-dependent results are reported as clinical performance.

The effect is not caused by model capacity: the architecture is identical in both runs, and the model is small enough that memorising general morphology is implausible. What differs is whether the test set contains beats from patients seen during training. Under Protocol A the model can rely on patient-specific morphology; under Protocol B it cannot, and the minority classes, whose appearance varies most between individuals, are the first to fail.

For applied work the implication is narrow but practical: a system validated only under beat-level splitting has not been shown to work on a new patient, regardless of the accuracy it reports. In the deployment that motivated the original model, every prediction was referred to a clinician for verification. On this evidence such a step is not a conservative design choice but a necessary one.

5. Limitations

This is a replication of a documented effect [3,4,6], not a new finding, and it should be read as confirmation rather than discovery. One dataset and one small architecture are used; the magnitude of the gap is specific to this pipeline, although its direction is consistent with published comparisons [4]. No class weighting, resampling or hyperparameter tuning is applied under either protocol, so the results do not establish how much minority-class performance could be recovered by rebalancing alone — a control worth adding. Five seeds are reported. Variation between seeds is small relative to the difference between protocols, but a larger number of runs would better characterise the tails. The beat-image rendering (window length, image size, line width) reproduces the original pipeline's representation but was not itself varied, and results may depend on those choices.

6. Conclusion

Holding a small image-based ECG classifier fixed and changing only the data-splitting protocol moves macro-F1 from 0.806 to 0.422 while overall accuracy moves only from 97.2 to 91.3 per cent. The accuracy figure conceals a near-total failure to detect supraventricular and fusion beats. This reproduces a known result on a new pipeline and supports existing recommendations that heartbeat classifiers be evaluated inter-patient and reported per class. Code and dataset construction are released in full.

References

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Reproducibility. Code for dataset construction, training and evaluation is released at github.com/OshiniWathsala1999/ecg-split-replication under an MIT licence. All figures reported here were produced with the PyTorch implementation in that repository, from an official PhysioNet download of the MIT-BIH Arrhythmia Database v1.0.0; an equivalent JAX implementation is released alongside it.